Chandrasekhar, dissatisfied with this Press—Teukolsky conclusion because of its crucial reliance on computer calculations, refused to concede our bet. Only when the entire proof could be done directly with formulas would he be fully convinced. Fifteen years later Bernard Whiting, a former postdoc of Hawking's (and thus an intellectual grandson of Sciama's), gave such a proof, and Chandrasekhar threw in the towe1.[7]

Chandrasekhar is even more of a perfectionist than 1. He and Zel'-dovich are at opposite ends of the perfectionist spectrum. So in 1975, when the youths of the golden age declared the golden age finished and exited from black-hole research en masse, Chandrasekhar was annoyed. These youths had carried Teukolsky's perturbation methods far enough to prove that black holes are probably stable, but they had not brought the methods into a form where other physicists could automatically compute *all* details of *any* desired black-hole perturbation—be it a pulsation, the gravitational waves from an infalling neutron star, a black-hole bomb, or whatever. This incompleteness was rankling.

Thus Chandrasekhar, in 1975 at age sixty-five, turned the full force of his mathematical prowess onto Teukolsky's equations. With unfailing energy and mathematical insight, he drove forward, through the complex mathematics, organizing it into a form that has been characterized as "rococo: splendorous, joyful, and immensely ornate." Finally in 1983, at age seventy-three, he completed his task and published a treatise entitled *The Mathematical Theory of Black Holes*—a treatise that will be a mathematical handbook for black-hole researchers for decades to come, a handbook from which they can extract methods for solving any black-hole perturbation problem that catches their fancy.

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1. As Saul Teukolsky, a compatriot of Bill Press's, recalls it, "This conference was Bill's response to what he considered a provocation. There was another conference going on, to which none of us had been invited. But all the gray eminences were attending, so Bill decided to have a conference only for young people."

2. In the late 1980s, at my mother's suggestion, the entire family requested excommunication from the Mormon Church in response to the Church's suppression of the rights of women.

3. This idea, while correct, has not yet produced any big payoffs, so I shall not discuss it in this book.

4. I have seen it unleashed in public only once. In 1971, on the occasion of his sixtieth birthday, Wheeler happened to be at an elegant banquet in a castle in Copenhagen—a banquet in honor of an